This is my *first* document prepared in LaTeX.

$$\begin{aligned} x + y - z &= 1 \\ x - y + z &= 1 \\ x + y + z &= 1 \end{aligned}$$

can be written in matrix terms as

$$\begin{pmatrix} 1 & 1 & -1 \\ 1 & -1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

Here, the matrix $\begin{pmatrix} 1 & 1 & -1 \\ 1 & -1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$ is invertible.

ldfkj A $\begin{pmatrix} 4 & 4 & 3 & 2 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix}$

B $\begin{pmatrix} 4 & 4 & 0 & 3 \\ 3 & 0 & 2 & 0 \\ 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 2 \end{pmatrix}$

T, subtable 0 $\begin{pmatrix} 0 & 0 & 0 & 0 \\ 4 & 4 & 0 & 3 \\ 3 & 3 & 0 & 1 \\ 2 & 2 & 0 & 4 \\ 1 & 1 & 0 & 2 \end{pmatrix}$

subtable 1 $\begin{pmatrix} 0 & 0 & 0 & 0 \\ 3 & 0 & 2 & 0 \\ 1 & 0 & 4 & 0 \\ 4 & 0 & 1 & 0 \\ 2 & 0 & 3 & 0 \end{pmatrix}$

subtable 2 $\begin{pmatrix} 0 & 0 & 0 & 0 \\ 3 & 1 & 3 & 3 \\ 1 & 2 & 1 & 1 \\ 4 & 3 & 4 & 4 \\ 2 & 4 & 2 & 2 \end{pmatrix}$

subtable 3 $\begin{pmatrix} 0 & 0 & 0 & 0 \\ 3 & 4 & 0 & 2 \\ 1 & 3 & 0 & 4 \\ 4 & 2 & 0 & 1 \\ 2 & 1 & 0 & 3 \end{pmatrix}$

$$\mathrm{C}\begin{pmatrix}3&2&2&0\\1&0&2&1\\4&1&3&1\\2&4&4&4\end{pmatrix}$$

$$\begin{bmatrix}3&1&-3&0&4&1\\0&1&-1&2&-2&5\\-2&-3&1&1&0&1\end{bmatrix}$$

$$\mathrm{A}\begin{pmatrix}4&4&3&2\\1&4&3&2\\3&2&3&4\\2&1&4&3\end{pmatrix}$$

$$\mathrm{T0}\begin{pmatrix}0&0&0&0\\4&4&0&3\\3&3&0&1\\2&2&0&4\\1&1&0&2\end{pmatrix}$$

$$\mathrm{T1}\begin{pmatrix}0&0&0&0\\3&0&2&0\\1&0&4&0\\4&0&1&0\\2&0&3&0\end{pmatrix}$$

$$\mathrm{T2}\begin{pmatrix}0&0&0&0\\3&1&3&3\\1&2&1&1\\4&3&4&4\\2&4&2&2\end{pmatrix}$$

$$\mathrm{T3}\begin{pmatrix}0&0&0&0\\3&4&0&2\\1&3&0&4\\4&2&0&1\\2&1&0&3\end{pmatrix}$$

$$\begin{pmatrix}3&3&0&1\end{pmatrix}+\begin{pmatrix}3&0&2&0\end{pmatrix}+\begin{pmatrix}2&4&2&2\end{pmatrix}+\begin{pmatrix}4&2&0&1\end{pmatrix}=\begin{pmatrix}2&4&4&4\end{pmatrix}.$$

C $\begin{pmatrix} 3 & 2 & 2 & 0 \\ 1 & 0 & 2 & 1 \\ 4 & 1 & 3 & 1 \\ 2 & 4 & 4 & 4 \end{pmatrix}$

GF2

A $\begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \end{pmatrix}$

B $\begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

T0 $\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}$

T1 $\begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}$

C $\begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 \end{pmatrix}$

B $\begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

T0 $\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}$

$$\mathrm{T1}\begin{pmatrix}0&0&0&0\\1&0&1&1\\0&0&0&1\\1&0&1&0\end{pmatrix}$$

$$\mathrm{T1}\begin{pmatrix}0&0&0&0\\0&1&0&0\\0&0&0&1\\1&0&1&0\end{pmatrix}$$

$$\mathrm{t}=2\begin{pmatrix}0&0&0&0\\4&1&1&3\\3&2&2&1\\2&3&3&4\\1&4&4&2\\4&0&3&3\\3&1&4&2\\2&2&0&4\\1&3&1&2\\0&4&2&0\\3&0&1&1\\2&1&2&4\\1&2&3&2\\0&3&4&0\\4&4&0&3\\2&0&4&4\\1&1&0&2\\0&2&1&0\\4&3&2&3\\3&4&3&1\\1&0&2&2\\0&1&3&0\\4&2&4&3\\3&3&0&1\\2&4&1&4\end{pmatrix}$$

$$\mathrm{t}=1\begin{pmatrix}0&0&0&0\\4&0&3&3\\3&0&1&1\\2&0&4&4\\1&0&2&2\end{pmatrix}$$

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 4 & 1 & 1 & 3 \\ 3 & 2 & 2 & 1 \\ 2 & 3 & 3 & 4 \\ 1 & 4 & 4 & 2 \end{pmatrix}$$